CT500 FirmWare Programming Guide

Version 0.3

Shmt Co., Ltd.

REVISION HISTORY PAGE

Issue	Rev No.	Originator	Details of Change	Date
1	0.1	DJKIM/BHLEE SHKIM/BF109UP	Initial Version	2007-03-01
2	0.2	SHKIM/ BF109UP	Modify pre driver	2007-03-02
3	0.3	BHLEE	Update DMA	2007-03-06

Technical documents for Firmware

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Contents

Figures

Tables

1. INTRODUCTION

본 문서는 CT500 의 firmware programming 시 사용 가능한 라이브러리 함수들에 대한내용을 다룬다.

2. DRIVERS FUNCTIONS

CT500 의 driver 함수는 pre/post-level driver 함수로 구성된다. Pre-level 드라이버 함수는 각 하드웨어 모듈의 레지스터 설정과 관련된 함수들이며, post-level 드라이버 함수들은 pre-level 드라이버를 이용한 각 하드웨어 모듈의 read/write와 같은 functional한 동작을 위한 드라이버 함수들이다.

3. BASIC PERIPHERALS

3.1. DMA

3.1.1. DMA2Init()

Description

DMA의 Schedule및 Resource를 초기화 한다.

시스템 시작과 동시에 한번만 실행을 해주면 된다.

```
void DMA2Init( smtBoolean polling)
```

Parameters

Polling

DMA동작 완료를 Polling으로 처리할지 인터럽트로 처리할지를 결정한다.

SMT_TRUE일 경우 Polling으로 처리하고 SMT_FALSE 일경우 인터럽트로 처리한다.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.1.2. smt2EDMACopy()

Description

Peripheral에서 memory로 DMA를 이용하여 데이터를 전송하는데 사용이 된다.

Memory에서 Memory로는 허용되지 않으며 그때는 smt2DMAMemCopy()를 사용해야 한다.

```
smtUint32 smt2EDMACopy(DMA_DEVICE device, EDMA_STRUCT *dma, smtUint8 descCount);
```

Parameters

device

EDMA에서는 미리정의된 peripheral에 따라 Channel이 정의되어 있다. 미리 정의된 device일 경우 해당하는 디바이스 값을 설정하고 그렇지 않을 경우 DMADEV_NONE를 사용한다.

dma

EDMA 설정값이 있다. Src, dst를 지정하고 각각의 bus width및 한번에 전송할 블록등을 설정한다.

descCount

디스크립터(DMA를 이용하여 전송하려는 내용을 여러 개를 디스크립터러 만들어서 한번에 적용하는데 사용)를 사용할경우 설정한 디스크립터 개수가 들어간다.

그렇지 않을경우 1 을 넣는다.(디스크립터를 사용하지 않음)

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
EDMA_STRUCT edma;


        edma.src        = (smtUint32)&NANDDATA;
        edma.dst        = (smtUint32)ppagebuf;
        edma.nBytes     = sizeInByte;

        edma.m2m        = SMT_FALSE;
        edma.srcWidth   = transSize;
        edma.dstWidth   = DMA_BUSWIDTH32;
        edma.tSize      = burstSize;
```

```
edma.srcInc = DMA_ADDR_NOINC;
edma.dstInc = DMA_ADDR_INC;

edma.startIntEn  = SMT_FALSE;
edma.EndIntEn    = SMT_FALSE;

edma.intFunction = &NANDDMAHandler;

smt2EDMACopy(DMADEV_NAND, &descriptor, 1);
```

3.1.3. smt2GDMACopy()

Description

2D 메모리영역을 2D 메모리 영역으로 복사하는데 사용이 된다. 이미지 복사에 사용이 된다.

```
smtUint32 smt2GDMACopy(GDMA_STRUCT *gdma);
```

Parameters

gdma

source 이미지의 크기(width, height)와 destination 이미지의 크기와 복사할 영역(sClipX, sClipY)와 복사될 영역(dClipX, dClipY)가 필요하다.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
gdma.src = (smtUint32)src_addr;
gdma.sWidth = 16;
gdma.sHeight = 16;   // source image 크기

gdma.sClipX = 2; // 복사할 영역의 시작 좌표
gdma.sClipY = 2;


gdma.dst = (smtUint32)dst_addr;
gdma.dWidth = 16;
gdma.dHeight = 16;// destination image 크기

gdma.dClipX = 4;
gdma.dClipY = 2; // 복사될 영역의 시작 좌표

gdma.clipWidth = 8;
gdma.clipHeight = 8; // 복사할 이미지 영역

gdma.bpp = 1;   // Bit Per Pixel
gdma.intFunction = &GDMAHandler; // 복사 완료후 수행할 루틴.

errCode = smt2GDMACopy(&gdma);
```

3.1.4. smt2DMAMemCopy ()

Description

DMA를 이용한 Memcpy 함수.

```
smtUint32 smt2DMAMemCopy(smtUint32 *src, smtUint32 *dst, smtUint32 length)
```

Parameters

src

source 메모리 영역

dst

target 메모리 영역

length

byte 단위 복사할 Byte

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
None
```

3.2. GPIO

3.2.1. smtGPIOSetDir()/smtGPIOGetDir()

Description

GPIO GPIO0_OE, GPIO1_OE register's software interface, set direction(in/out/bidirection) of GPIO

```
void smtGPIOSetDir(GPIO_STRUCT gpio);
void smtGPIOGetDir(GPIO_STRUCT gpio);
```

Parameters

gpio

refer to GPIO0_TYPE structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.2.2. smtGPIOSetData()/smtGPIOGetData()

Description

GPIO GPIO0_IN, GPIO0_OUT, GPIO1_IN, GPIO1_OUT register's software interface, write data or read data from GPIO controller

```
void smtGPIOSetData(GPIO_STRUCT gpio, smtUint32 gpioData);
void smtGPIOGetData(GPIO_STRUCT gpio, smtUint32 gpioData);
```

Parameters

gpio

refer to GPIO0_TYPE structure

gpioData

data write to GPIO, read from GPIO

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
None
```

3.2.3. smtGPIOIntClear()

Description

GPIO0_INTSTAT, GPIO1_INTSTAT register's software interface, clear GPIO interrupt pending register

```
void smtGPIOIntClear(GPIO_STRUCT gpio);
```

Parameters

gpio

refer to GPIO0_TYPE structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.2.4. smtGPIOIntEnable()/smtGPIOIntDisable()

Description

GPIO0_INTEN, GPIO1_INTEN register's software interface, make enable of disable interrupt of GPIO

```
void smtGPIOIntEnable(GPIO_STRUCT gpio);
void smtGPIOIntDisable(GPIO_STRUCT gpio);
```

Parameters

gpio

refer to GPIO0_TYPE structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.2.5. smtGPIOSetIntProperty()/smtGPIOGetIntProperty()

Description

GPIO0_INTLEVEL, GPIO0_INTPOL, GPIO0_INTBEDGE, GPIO0_INTLEVEL, GPIO0_INTPOL, GPIO0_INTBEDGE, register's software interface. Set GPIO's interrupt level, interrupt polarity, interrupt trigger mode.

```
void smtGPIOSetIntProperty(GPIO_STRUCT gpio);
void smtGPIOGetIntProperty(GPIO_STRUCT gpio);
```

Parameters

gpio

refer to GPIO0_TYPE structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.2.6. smt2GPIOWriteData()/smt2GPIOReadData()

Description

Write data or Read data form GPIO0/1

```
smtBoolean smt2GPIOWriteData(GPIO_STRUCT gpio, smtUint32 gpioData);
smtBoolean smt2GPIOReadData(GPIO_STRUCT gpio, smtUint32 gpioData);
```

Parameters

gpio

refer to GPIO0_TYPE structure

gpioData

refer to GPIO0_TYPE structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lib.h

Source: lib.c

Library: none

Example Code

```
none
```

3.3. VIC

3.3.1. EnableVIC()/DisableVIC()

Description

Write data or Read data form GPIO0/1

```
void EnableVIC(smtUint32 Polarity, smtUint32 Level, smtUint32 IntMod);
void DisableVIC(void);
```

Parameters

Polarity

Polarity of 32 interrupt source

Level

Interrupt event mode, level, edge mode

IntMod

Interrupt mode, IRQ/FIQ mode

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.3.2. RequestIRQ()/ReleaseIRQ()

Description

Write data or Read data form GPIO0/1

```
void RequestIRQ(smtUint32 IRQ, ISRType ISR);
void ReleaseIRQ(smtUint32 IRQ);
```

Parameters

IRQ

IRQ number to Requested or Released

ISR

Interrupt handler to registry to VIC function table

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.3.3. RequestIRQ()/ReleaseIRQ()

Description

Write data or Read data form GPIO0/1

```
void RequestIRQ(smtUint32 IRQ, ISRType ISR);
void ReleaseIRQ(smtUint32 IRQ);
```

Parameters

IRQ

IRQ number to Requested or Released

ISR

Interrupt handler to registry to VIC function table

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.3.4. EnableIRQ()/DisableIRQ()

Description

Enable/Disable IRQ of VIC

```
void EnableIRQ(smtUint32 IRQ);
void DisableIRQ(smtUint32 IRQ);
```

Parameters

IRQ

IRQ number to enable or disable

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.3.5. AckIRQ()

Description

Clear interrupt source pending register for edge interrupt

```
void AckIRQ(smtUint32 IRQ)
```

Parameters

IRQ

IRQ number to enable or disable

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.3.6. InterruptHandler()

Description

System IRQ top interrupt handler

```
void InterruptHandler(void);
```

Parameters

none.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

3.3.7. UndefHandler()

Description

System undefined instruction handler

```
void UndefHandler(int IRQ);
```

Parameters

IRQ

IRQ number to enable or disable

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: irq.h

Source: irq.c

Library: none

Example Code

```
none
```

4. COMMUNICATION DEVICES

4.1. I2C

4.1.1. smtI2CSetCtrl()

Description

이 함수는 I2C 통신을 위한 clk, operation mode, interrupt/ack enable 등을 설정한다.

```
void smtI2CSetCtrl (
    I2C_CTRL_STRUCT i2cCtrl
);
```

Parameters

i2cCtrl

I2c 설정 구조체.

Return Value

void.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.2. smtI2CGetCtrl()

Description

이 함수는 I2C의 clk, operation mode, interrupt/ack enable 등 정보를 가져온다.

```
Void smtI2CGetCtrl (
     I2C_CTRL_STRUCT i2cCtrl
);
```

Parameters

i2cCtrl

I2c 설정 구조체.

Return Value

void.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.3. smtI2CGetStatus()

Description

이 함수는 I2C의 status 레지스터 정보를 가져온다.

```
void smtI2CGetStatus (
    I2C_STATUS_STRUCT i2cStat
);
```

Parameters

i2cStat

I2c 설정 구조체.

Return Value

void.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.4. smtI2CSWReset()

Description

이 함수는 I2C soft reset한다.

```
void smtI2CSWReset (
     void
);
```

Parameters

none

Return Value

void.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.5. smtI2CEnable()

Description

이 함수는 I2C의 동작을 en/disable한다.

```
void smtI2CEnable (
    smtUint8 enable
);
```

Parameters

enable

Return Value

void.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.6. smtI2CInterruptWait()

Description

이 함수는 I2C의 pending flag가 발생하기를 기다린다.

```
void smtI2CInterruptWait (
     void
);
```

Parameters

void.

Return Value

void.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.7. smtI2CInterruptPendingClear()

Description

이 함수는 I2C의 pending flag를 clear 한다.

```
void smtI2CInterruptPendingClear (
     void
);
```

Parameters

void

Return Value

void.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.8. smtI2CInterruptPended()

Description

이 함수는 I2C pending flag를 return 한다.

```
smtInt32 smtI2CInterruptPended (
     void
);
```

Parameters

void

Return Value

void.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.9. smtI2CWrite()

Description

이 함수는 I2C를 통해 1byte 의 데이터 전송.

```
smtUint32smt2I2CWrite(
    smtUint8 addr,
    smtUint8 subAddr,
    smtUint8 data
);
```

Parameters

addr

slave device address.

subAddr

slave device subaddress.

data

전송할 데이터.

Return Value

SMT_SUCCESS/ SMT_ERROR

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.10. smtI2CRead()

Description

이 함수는 I2C를 통해 slave device에서 데이터를 읽어들인다.

```
smtUint32smtI2CRead (
      smtUint8 addr,
      smtUint8 subAddr,
      smtUint8 *data
);
```

Parameters

addr

slave device address.

subAddr

slave device subaddress.

data

읽어들인 데이터 저장.

Return Value

SMT_SUCCESS/ SMT_ERROR

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.1.11. smtI2CInit()

Description

이 함수는 I2C를 초기화한다.

```
void smtI2CInit (
    smtUint16 prescaler
);
```

Parameters

prescaler

I2C prescale value.

Return Value

void

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2c_pre_drv.h

Source: i2c_pre_drv.c

Library: none

Example Code

```
None.
```

4.2. I2S

4.2.1. smtSetClockMode()/smtGetClockMode()

Description

I2S_CLKCTRL 의 software interface

```
void smtSetClockMode(I2S_CLOCK_MODE *pclkMode)
void smtGetClockMode(I2S_CLOCK_MODE *pclkMode)
```

Parameters

pclkMode

I2S_CLKCTRL register's 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\comm\i2s.c)
```

4.2.2. smtI2SFIFOClear()

Description

I2S_STATUS RX/TX status clear bis's software interface

```
void smtI2SFIFOClear(I2S_TRANS_DIR dir)
```

Parameters

dir

I2S_DIR_RX is I2S RX FIFO clear, I2S_DIR_TX is I2S TX FIFO clear,

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\comm\i2s.c)
```

none.

4.2.3. smtI2SFIFOReset()

Description

I2S_CTRL의 RX/TX reset bit software interface

```
void smtI2SFIFOReset(I2S_TRANS_DIR dir)
```

Parameters

dir

I2S_DIR_RX is I2S RX FIFO reset, I2S_DIR_TX is I2S TX FIFO reset,

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\comm\i2s.c)
```

4.2.4. smtSetI2SMode()/smtGetI2SMode()

Description

I2S_CTRL's mode bits software interface

```
void smtSetI2SMode(I2S_TRANS_DIR dir, I2S_CONTROL *pcontrol)
```

Parameters

dir

I2S_DIR_RX is I2S RX FIFO reset, I2S_DIR_TX is I2S TX FIFO reset,

pcontrol

I2S_CTRL's DMAIntEn, FifoOURIntEn, WLen, Ljust, FifoTH 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\comm\i2s.c)
```

4.2.5. smtI2SEnable()

Description

I2S_CTRL's RX/TX enable bit software interface

```
void smtI2SEnable(I2S_TRANS_DIR dir, smtUint32 enable)
```

Parameters

dir

Direction of I2S interface, I2S_DIR_RX is RX mode, I2S_DIR_TXs TX mode.

enable

true is enable, false is disable

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\comm\i2s.c)
```

4.2.6. smtI2SGetStatus()

Description

I2S_STATUS's RX/TX status bit software interface

```
void smtI2SGetStatus(I2S_TRANS_DIR dir, I2S_STATUS_DAT *pstatus)
```

Parameters

dir

Direction of I2S interface, I2S_DIR_RX is RX mode, I2S_DIR_TXs TX mode.

pstatus

I2S_STATUS register 1:1 mapping layer per RX/TX

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\comm\i2s.c)
```

4.2.7. smtI2SSetData()/smtI2SGetData()

Description

I2S_STATUS's RX/TX status bit software interface

```
void smtI2SSetData(smtUint32 data)
smtUint32          smtI2SGetData(void)
```

Parameters

data

write data to I2S_DATA register

Return Value

Read data from I2S_DATA register.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: i2s_pre_drv.h

Source: i2s_pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\comm\i2s.c)
```

4.3. SPI

4.3.1. smtSPISetMode()

Description

이 함수는 SPI의 모드를 설정한다.

```
void smtSPISetMode (
    smtUint8 channel,
    SPI_CON_STRUCT spiCtrl
);
```

Parameters

channel

spi channel selection.

spiCtrl

SPI control 구조체

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.2. smtSPIGetMode()

Description

이 함수는 SPI의 모드 정보를 읽어온다.

```
void smtSPIGetMode (
     smtUint8 channel,
     SPI_CON_STRUCT spiCtrl
);
```

Parameters

channel

spi channel selection.

spiCtrl

SPI control 구조체

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.3. smtSPISetMode()

Description

이 함수는 SPI의 speed(prescale)를 설정한다.

```
void smtSPISetPrescaler (
     smtUint8 channel,
     smtUint32 preVal,
     smtUint32 divVal
);
```

Parameters

channel

spi channel selection.

preVal

SPI prescaler value

divVal

SPI divider value

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.4. smtSPIGetStatus()

Description

이 함수는 SPI의 상태 정보를 읽어온다.

```
void smtSPIGetStatus (
    smtUint8 channel,
    SPI_STATUS_STRUCT spiStatus
);
```

Parameters

channel

spi channel selection.

spiStatus

SPI 상태 정보 구조체

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.5. smtSPISetIntDMA()

Description

이 함수는 SPI의 interrupt en/disable 및 DMA en/disable 등을 설정한다.

```
void smtSPISetIntDMA (
      smtUint8 channel,
      SPI_INTDMA_STRUCT spiIntDMA
);
```

Parameters

channel

spi channel selection.

spiIntDMA

SPI interrupt, DMA 설정 구조체

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.6. smtSPIGetIntDMA()

Description

이 함수는 SPI의 interrupt, DMA 설정 정보를 읽어온다.

```
void smtSPIGetIntDMA (
    smtUint8 channel,
    SPI_INTDMA_STRUCT spiIntDMA
);
```

Parameters

channel

spi channel selection.

spiIntDMA

SPI interrupt, DMA 설정 구조체

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.7. smtSPIGetIntStatus()

Description

이 함수는 SPI의 인터럽트 정보를 읽어온다.

```
void smtSPIGetIntStatus (
    smtUint8 channel,
    SPI_INTSTA_STRUCT spitIntDMA
);
```

Parameters

channel

spi channel selection.

spiIntDMA

SPI 인터럽트 상태 구조체

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.8. smtSPIClrIntStatus()

Description

이 함수는 SPI의 인터럽트 상태를 clear 한다.

```
void smtSPIClrIntStatus (
    smtUint8 channel,
    SPI_INTSTA_STRUCT spitIntDMA
);
```

Parameters

channel

spi channel selection.

spiIntDMA

SPI 인터럽트 상태 구조체

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.9. smtSPIOnebyteWrite()

Description

이 함수는 SPI를 통해 1byte 전송한다.

```
void smtSPIOnebyteWrite (
    smtInt8 channel,
    smtInt8 data
);
```

Parameters

channel

spi channel selection.

data

전송할 1 바이트 데이터

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.10. smtSPIOnebyteRead()

Description

이 함수는 SPI를 통해 1byte를 읽어온다.

```
smtUint8 smtSPIOnebyteRead (
    smtInt8 channel
);
```

Parameters

channel

spi channel selection.

Return Value

smtUint8

읽어온 1 바이트 데이터

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.3.11. smtSPIInit()

Description

이 함수는 SPI를 통해 1byte 전송한다.

```
void smtSPIInit (
    smtInt8 channel,
    smtUint32 PreVal,
    smtUint32 DivVal
);
```

Parameters

channel

spi channel selection.

PreVal

SPI prescaler value.

DivVal

SPI divider value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: spi_pre_drv.h

Source: spi_pre_drv.c

Library: none

Example Code

```
None.
```

4.4. UART

4.4.1. smt2UartInit()

Description

UART 초기화.

```
smtBoolean
smt2UartInit(
    UartConfig uartCfg
);
```

Parameters

uartCfg

UART 초기화 정보를 담고 있는 구조체.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.2. smt2UartDeInit ()

Description

initialize 한 UART 채널을 disable.

```
void
smt2UartDeInit(
    void
);
```

Parameters

none.

Return Value

none

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none
```

4.4.3. smt2UartPutStr()

Description

UART를 통해 문자열을 출력.

```
smtBoolean
smt2UartPutStr(
    smtInt8 *pData
);
```

Parameters

pData

uart를 통해 출력할 문자열.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

입력 문자열의 마지막은 '\0' 여야 한다.

Requirement

Platform: Requires CT500 firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.4. smt2UartGetStr()

Description

UART를 통해 문자열을 입력 받는다.

```
smtBoolean
smt2UartGetStr(
    smtUint8 *pData,
    smtUint8 dataCnt
);
```

Parameters

pData

uart를 통해 출력할 문자열.

dataCnt

읽어들일 문자열의 개수.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.5. smt2UartPrint()

Description

UART를 통해 포맷 형식의 문자열을 출력한다.

```
smtBoolean
smt2UartPrint(
    int dispLvl,
    char *fmt,
    [argument]...
);
```

Parameters

dispLvl

출력 레벨 설정값.

fmt

출력 포맷 설정.

[argument]...

선택적 인자들.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.6. smt2UartPutCh()

Description

UART를 통해 한 개의 문자를 출력한다.

```
smtBoolean
smt2UartPutCh(
    smtInt8 data
);
```

Parameters

data

출력할 문자 데이터.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.7. smt2UartGetCh()

Description

UART를 통해 한 개의 문자를 입력 받는다.

```
smtUint8
smt2UartGetCh(
    smtUint32 waitTime
);
```

Parameters

waitTime

time out value.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: uart_post_drv.h

Source: uart_post_drv.c

Library: none

Example Code

```
none.
```

4.4.8. smtUartSetMode()

Description

UART master/status/baud generation/RX timeout register 변경.

```
smtBoolean
smtUartSetMode(
    UartChannel uartCh,
    UartStruct *pUart,
    UartMode mode
);
```

Parameters

uartCh

UART channel 선택.

pUart

register setting 정보를 담고 있는 구조체 포인터.

mode

setting 할 target register를 선택.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

4.4.9. smtUartGetMode()

Description

UART master/status/baud generation/RX timeout register 값을 읽어서 인수로 넘어온 구조체에 저장한다..

```
smtBoolean
smtUartGetMode(
     UartChannel uartCh,
     UartStruct *pUart,
     UartMode mode
);
```

Parameters

uartCh

UART channel 선택.

pUart

register setting 정보를 저장할 구조체 포인터.

mode

읽어들일 target register를 선택.

Return Value

SMT_SUCCESS 이면 수행 성공

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

4.4.10. smtUartReadRxFifo()

Description

UART RX FIFO에서 1byte size의 데이터를 읽어 온다.

```
smtUint8
smtUartReadRxFifo(
    UartChannel uartCh
);
```

Parameters

uartCh

UART channel 선택.

Return Value

RX FIFO에 읽어 들인 1byte size의 데이터.

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

4.4.11. smtUartWriteTxFifo()

Description

UART RX FIFO에서 1byte size의 데이터를 읽어 온다.

```
void
smtUartWriteTxFifo(
    UartChannel uartCh,
    smtUint8 txData
);
```

Parameters

uartCh

UART channel 선택.

txData

TX FIFO에 쓸 데이터.

Return Value

none.

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

4.4.12. smtUartDataAvailable()

Description

UART RX FIFO의 count 값을 체크 하여 읽어 들일 데이터가 있는지를 검사한다.

```
smtUint32 smtUartDataAvailable(void);
```

Parameters

none.

Return Value

none.

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: uart_pre_drv.h

Source: uart_pre_drv.c

Library: none

Example Code

```
none.
```

4.5. USB

Not yet

5. DISPLAY DEVICES

5.1. DM

5.1.1. smtDMSetMasterCtrl()

Description

DM(Display Module)의 DMCON register를 설정한다.

```
void
smtDMSetMasterCtrl(
    DMCtrl maCtrl
);
```

Parameters

maCtrl

DMCON register wrapper 구조체.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMCtrl dmCtrl;

//1. set DM control
dmCtrl.prioritySel        = 0x1;// priority select
dmCtrl.swReset            = 0x0;// DM software reset
dmCtrl.errIntEn           = 0x0;// enable DM error interrupt
dmCtrl.intEn              = 0x1;// enable frame start interrupt
smtDMSetMasterCtrl(dmCtrl);
```

5.1.2. smtDMGetMasterCtrl()

Description

DM(Display Module)의 DMCON register를 읽어 들인다.

```
void
smtDMGetMasterCtrl(
    DMCtrl *pMaCtrl
);
```

Parameters

pMaCtrl

DMCON register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMCtrl dmMasterCtrl;
smtDMGetMasterCtrl(&dmMasterCtrl);
dmMasterCtrl.swReset =0x1;
smtDMSetMasterCtrl(dmMasterCtrl);
```

5.1.3. smtDMGetMasterStatus()

Description

DM(Display Module)의 LCDMSTS register를 읽어 들인다.

```
void
smtDMGetMasterStatus(
    DMStatus *pMaStat
);
```

Parameters

pMaStat

LCDMSTS register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMStatus dmSts;
smtDMGetMasterStatus(&dmSts);//read DM status register

if(dmSts.startFrame == 1)
{
        if(dmSts.evenFieldInt == 1)
                bChgOddFrame = SMT_TRUE;
        else
                bChgEvenFrame = SMT_TRUE;
}
```

5.1.4. smtDMSetCCtrl()/smtDMGetCCtrl()

Description

DM(Display Module)의 cursor plane control register를 read/write.

```
void smtDMSetCCtrl(DMPlaneCtrl Ctrl);
void smtDMGetCCtrl(DMPlaneCtrl *pCtrl);
```

Parameters

ctrl

cursor plane control register wrapper 구조체.

pCtrl

cursor plane control register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
//4. set plane control register
cPlaneCfg.xEn           = 0x1;
cPlaneCfg.pixFmt        = 0x3;//0:8BPP,1:RGB332 2:ARGB(1555) 3:RGB565
cPlaneCfg.gammaEn       = 0x0;
cPlaneCfg.sWidth        = CURSOR_WIDTH;
cPlaneCfg.sHeight       = CURSOR_HEIGHT;
smtDMSetCCtrl(cPlaneCfg);
```

5.1.5. smtDMSetCBlndMode()/smtDMGetCBlndMode()

Description

DM(Display Module)의 cursor plane blend mode register를 read/write.

```
void smtDMSetCBlndMode(DMPlaneBlndMode blndMode);
void smtDMGetCBlndMode(DMPlaneBlndMode *pBlndMode);
```

Parameters

blndMode

cursor plane blend mode register wrapper 구조체.

pBlndMode

cursor plane blend mode register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
cPlaneBlndMode.xPos = 0;
cPlaneBlndMode.yPos = 0;
cPlaneBlndMode.colKeyEn = 0x0; //dis/enable color key
cPlaneBlndMode.blendMod = 0x0; //0:no alpha , 2: global alpha, 3: pixel alpha
smtDMSetCBlndMode(cPlaneBlndMode);
```

5.1.6. smtDMSetCBlnd()/smtDMGetCBlnd()

Description

DM(Display Module)의 cursor plane blend register를 read/write.

```
void smtDMSetCBlnd(DMPlaneBlnd blnd);
void smtDMGetCBlnd(DMPlaneBlnd *pblnd);
```

Parameters

blnd

cursor plane blend register wrapper 구조체.

pBlnd

cursor plane blend register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
//1. set chromakey/alpha value(BLEND)
cPlaneBlnd.alpha = 0xFF;// global alpha vale
cPlaneBlnd.colKey          = 0xFFFFFF;
smtDMSetCBlnd(cPlaneBlnd);
```

5.1.7. smtDMSetCBase()/smtDMGetCBase()

Description

DM(Display Module)의 cursor plane base address register를 read/write.

```
void smtDMSetCBase(smtUint16 planeXRef);
smtUint16 smtDMGetCBase(void);
```

Parameters

planeXRef

cursor plane XRef value.

Return Value

none.

Remarks

plane source image의 width 와 plane의 width가 같은 경우 xRef는 0 이 된다.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
smtDMSetCBase(0x0);
```

5.1.8. smtDMSetCAddr()/smtDMGetCAddr()

Description

DM(Display Module)의 cursor plane start address register를 read/write.

```
void smtDMSetGAddr(smtUint32 addr);
smtUint32 smtDMGetGAddr(void);
```

Parameters

addr

cursor plane start addr value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
smtUint16 *frame0 = (smtUint16 *)CURSOR_BASEADDR;
smtDMSetCAddr((smtUint32)frame0);
```

5.1.9. smtDMSetCPalette()/smtDMGetCPalette()

Description

DM(Display Module)의 cursor plane palette register를 read/write.

```
void smtDMSetCPalette(DMPlanePalette palette);
void smtDMGetCPalette(DMPlanePalette *pPalette);
```

Parameters

palette

cursor plane palette register 정보를 담고 있는 구조체.

pPalette

cursor plane palette register 정보를 담고 있는 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
none.
```

5.1.10. smtDMSetGCtrl()/smtDMGetGCtrl()

Description

DM(Display Module)의 graphic plane control register를 read/write.

```
void smtDMSetGCtrl(DMPlaneCtrl ctrl);
void smtDMGetGCtrl(DMPlaneCtrl *pCtrl);
```

Parameters

ctrl

graphic plane control register wrapper 구조체.

pCtrl

graphic plane control register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMPlaneCtrl gPlaneCfg;

gPlaneCfg.xEn           = 0x1;
gPlaneCfg.pixFmt        = 0x4;//0:8BPP,2:RGB565,3:ARGB1555, 4:RGB888, 5:ARGB8888
gPlaneCfg.gammaEn       = 0x0;
gPlaneCfg.sWidth        = GRAPHIC_WIDTH;
gPlaneCfg.sHeight       = GRAPHIC_HEIGHT;

smtDMSetGCtrl(gPlaneCfg);
```

5.1.11. smtDMSetGBlndMode()/smtDMGetGBlndMode()

Description

DM(Display Module)의 graphic plane blend mode register를 read/write.

```
void smtDMSetGBlndMode(DMPlaneBlndMode blndMode);
void smtDMGetGBlndMode(DMPlaneBlndMode *pBlndMode);
```

Parameters

blndMode

graphic plane blend mode register wrapper 구조체.

pBlndMode

graphic plane blend mode register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMPlaneBlndMode gPlaneBlndMode;
gPlaneBlndMode.xPos = 0;
gPlaneBlndMode.yPos = 0;
gPlaneBlndMode.colKeyEn = 0x0;
gPlaneBlndMode.blendMod = 0x2; //0:no alpha , 2: global alpha, 3: pixel alpha
smtDMSetGBlndMode(gPlaneBlndMode);
```

5.1.12. smtDMSetGBlnd()/smtDMGetGBlnd()

Description

DM(Display Module)의 graphic plane blend register를 read/write.

```
void smtDMSetCBlnd(DMPlaneBlnd blnd);
void smtDMGetCBlnd(DMPlaneBlnd *pblnd);
```

Parameters

blnd

graphic plane blend register wrapper 구조체.

pBlnd

graphic plane blend register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
gPlaneBlnd.alpha = 0xFF;
gPlaneBlnd.colKey           = 0xFFFFFF;
smtDMSetGBlnd(gPlaneBlnd);
```

5.1.13. smtDMSetGBase()/smtDMGetGBase()

Description

DM(Display Module)의 graphic plane base address register를 read/write.

```
void smtDMSetCBase(smtUint16 planeXRef);
smtUint16 smtDMGetCBase(void);
```

Parameters

planeXRef

graphic plane XRef value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
smtDMSetGBase(0x0);
```

5.1.14. smtDMSetGAddr()/smtDMGetGAddr()

Description

DM(Display Module)의 graphic plane start address register를 read/write.

```
void smtDMSetCAddr(smtUint32 addr);
smtUint32 smtDMGetCAddr(void);
```

Parameters

addr

graphic plane start addr value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
smtUint32 *frame0 = (smtUint32 *)GRAPHIC_BASEADDR;
smtDMSetGAddr((smtUint32)frame0);
```

5.1.15. smtDMSetGPalette()/smtDMGetGPalette()

Description

DM(Display Module)의 graphic plane palette register를 read/write.

```
void smtDMSetCPalette(DMPlanePalette palette);
void smtDMGetCPalette(DMPlanePalette *pPalette);
```

Parameters

palette

graphic plane palette register wrapper 구조체.

pPalette

graphic plane palette register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
none.
```

5.1.16. smtDMSetGGamma()/smtDMGetGGamma()

Description

DM(Display Module)의 graphic plane gamma register를 read/write.

```
void smtDMSetGGamma(smtUint32 addr, smtUint32 gammaVal);
smtUint32 smtDMGetGGamma(smtUint32 addr);
```

Parameters

addr

Gamma look up table index.

gammaVal

Gamma look up table index에 해당하는 gamma value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
none.
```

5.1.17. smtDMSetVCtrl()/smtDMGetVCtrl()

Description

DM(Display Module)의 video plane control register를 read/write.

```
void smtDMSetVCtrl(DMPlaneCtrl ctrl);
void smtDMGetVCtrl(DMPlaneCtrl *pCtrl);
```

Parameters

ctrl

video plane control register wrapper 구조체.

pCtrl

video plane control register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
vPlaneCfg.xEn              = 0x1;
vPlaneCfg.pixFmt           = 0x3;//
vPlaneCfg.vPlaneIFEn = 0x0;// interlaced video source
vPlaneCfg.vPlaneN2PUpEn = 0x0;
vPlaneCfg.sWidth           = VIDEO_WIDTH;
vPlaneCfg.sHeight          = VIDEO_HEIGHT;
smtDMSetVCtrl(vPlaneCfg);
```

5.1.18. smtDMSetVBlndMode()/smtDMGetVBlndMode()

Description

DM(Display Module)의 video plane blend mode register를 read/write.

```
void smtDMSetVBlndMode(DMPlaneBlndMode blndMode);
void smtDMGetVBlndMode(DMPlaneBlndMode *pBlndMode);
```

Parameters

blndMode

video plane blend mode register wrapper 구조체.

pBlndMode

video plane blend mode register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
vPlaneBlndMode.xPos = 0;
vPlaneBlndMode.yPos = 0;
vPlaneBlndMode.colKeyEn = 0x0;
vPlaneBlndMode.blendMod = 0x0; //0:no alpha , 2: global alpha, 3: pixel alpha
smtDMSetVBlndMode(vPlaneBlndMode);
```

5.1.19. smtDMSetVBlnd()/smtDMGetVBlnd()

Description

DM(Display Module)의 video plane blend register를 read/write.

```
void smtDMSetVBlnd(DMPlaneBlnd blnd);
void smtDMGetVBlnd(DMPlaneBlnd *pblnd);
```

Parameters

blnd

video plane blend register wrapper 구조체.

pBlnd

video plane blend register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
vPlaneBlnd.alpha = 0xFF;
vPlaneBlnd.colKey           = 0xFFFFFF;
smtDMSetVBlnd(vPlaneBlnd);
```

5.1.20. smtDMSetVBase()/smtDMGetVBase()

Description

DM(Display Module)의 video plane base address register를 read/write.

```
void smtDMSetVBase(smtUint16 planeXRef);
smtUint16 smtDMGetVBase(void);
```

Parameters

planeXRef

video plane XRef value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
smtDMSetVBase(0x0);
```

5.1.21. smtDMSetVAddr()/smtDMGetGVddr()

Description

DM(Display Module)의 video plane start address register를 read/write.

```
void smtDMSetVAddr(smtUint32 addr);
smtUint32 smtDMGetVAddr(void);
```

Parameters

addr

video plane start addr value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
smtUint16 *frame0 = (smtUint16 *)VIDEO_BASEADDR;
smtDMSetVAddr((smtUint32)frame0);
```

5.1.22. smtDMSetVGamma()/smtDMGetVGamma()

Description

DM(Display Module)의 video plane gamma register를 read/write.

```
void smtDMSetVGamma(smtUint32 addr, smtUint32 gammaVal);
smtUint32 smtDMGetVGamma(smtUint32 addr);
```

Parameters

addr

Gamma look up table index.

gammaVal

Gamma look up table index에 해당하는 gamma value.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
none.
```

5.1.23. smtDMSetBGCol()/smtDMGetBGCol()

Description

DM(Display Module)의 back ground plane background color register를 read/write.

```
void smtDMSetBGCol(smtUint32 colour);
smtUint32 smtDMGetBGCol(void);
```

Parameters

colour

background plane의 색상 값.(RGB888 단색만 지원)

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
smtDMSetBGCol(0xFF0000);//set back ground color to red
```

5.1.24. smtDMSetLCDCtrl()/smtDMGetLCDCtrl()

Description

DM(Display Module)의 lcd control register를 read/write.

```
void smtDMSetLCDCtrl(DMLcdCtrl ctrl);
void smtDMGetLCDCtrl(DMLcdCtrl *pCtrl);
```

Parameters

ctrl

lcd control register wrapper 구조체.

pCtrl

lcd control register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMLcdCtrl dmLcdCtrl;

dmLcdCtrl.invertHSync              = LCD_IHS;
dmLcdCtrl.invertVSync              = LCD_IVS;
dmLcdCtrl.invertPixClk             = LCD_IPS;
dmLcdCtrl.invertWrEn               = LCD_IEO;
dmLcdCtrl.lcdBpp          = LCD_BPP;
dmLcdCtrl.rbSwap                   = LCD_BGR;
dmLcdCtrl.pDiv                     = 0x0;
dmLcdCtrl.lcdPwrEn                 = 0x1;
dmLcdCtrl.lcdEn           = LCD_VIDEO_SYNC_EN; // video sync enable

smtDMSetLCDCtrl(dmLcdCtrl);
```

5.1.25. smtDMSetHSync0()/smtDMGetHSync0()

Description

DM(Display Module)의 lcd horizontal sync0 register를 read/write.

```
void smtDMSetHSync0(DMHSync0Ctrl hSync0);
void smtDMGetHSync0(DMHSync0Ctrl *pHSync0);
```

Parameters

hSync0

lcd hSync0 register wrapper 구조체.

pHSync0

lcd hSync0 register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMHSync0Ctrl dmHSync0;
dmHSync0.tHFrontPorch           = LCD_HFP;
dmHSync0.tHBackPorch            = LCD_HBP;
smtDMSetHSync0(dmHSync0);
```

5.1.26. smtDMSetHSync1()/smtDMGetHSync1()

Description

DM(Display Module)의 lcd horizontal sync1 register를 read/write.

```
void smtDMSetHSync1(DMHSync0Ctrl hSync1);
void smtDMGetHSync1(DMHSync0Ctrl *pHSync1);
```

Parameters

hSync1

lcd hSync1 register wrapper 구조체.

pHSync1

lcd hSync1 register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMHSync1Ctrl dmHSync1;
dmHSync1.tHDspPixPerLine          = LCD_CPL;
dmHSync1.tHPulsWidth              = LCD_HSW;
smtDMSetHSync1(dmHSync1);
```

5.1.27. smtDMSetVSync0()/smtDMGetVSync0()

Description

DM(Display Module)의 lcd vertical sync0 register를 read/write.

```
void smtDMSetVSync0(DMVSync0Ctrl vSync0);
void smtDMGetVSync0(DMVSync0Ctrl *pVSync0);
```

Parameters

vSync0

lcd vSync0 register wrapper 구조체.

pVSync0

lcd vSync0 register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMVSync0Ctrl dmVSync0;

dmVSync0.tVFrontPorch           = LCD_VFP;
dmVSync0.tVBackPorch            = LCD_VBP;
smtDMSetVSync0(dmVSync0);
```

5.1.28. smtDMSetVSync1()/smtDMGetVSync1()

Description

DM(Display Module)의 lcd vertical sync1 register를 read/write.

```
void smtDMSetVSync1(DMVSync1Ctrl vSync1);
void smtDMGetVSync1(DMVSync1Ctrl *pVSync1);
```

Parameters

vSync1

lcd vSync1 register wrapper 구조체.

pVSync1

lcd vSync1 register wrapper 구조체 포인터.

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: lcd_pre_drv.h

Source: lcd_pre_drv.c

Library: none

Example Code

```
DMVSync1Ctrl dmVSync1;

dmVSync1.tVDspPeriod    = LCD_LPS;
dmVSync1.tVPulsWidth    = LCD_VSW;
smtDMSetVSync1(dmVSync1);
```

5.2. VIDEO ENCODER

5.2.1. smtSetVideoEncode()

Description

set video encoder의 control / internal control registers.

```
void
smtSetVideoEncode(
    VEncCtrl *vCtrl,
    VEncInternal *vInternal
);
```

Parameters

vCtrl

video encoder control register 정보를 담고 있는 구조체의 포인터.

vInternal

video encoder internal control register 정보를 담고 있는 구조체의 포인터.

Return Value

none.

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: videoenc_pre_drv.h

Source: videoenc_pre_drv.c

Library: none

Example Code

```
VEncInternal vInternal;
vInternal.hSyncWidth       = HSYNC_WID;
vInternal.burstWidth       = BURST_WID;
vInternal.lDelay  = LUMA_DELAY;
vInternal.cDelay  = CHRO_DELAY;
vInternal.patternMode      = EN_INTERNAL_PATTERN;
vInternal.pattern = COLOR_PATTERN_MODE;
vInternal.resetSCH         = EN_REST_SCH;
vInternal.color    = EN_COLOR_KILL;
vInternal.CFilter  = CHRO_FILTER_SEL;
vInternal.lFilter  = LUMA_FILTER_SEL;

smtSetVideoEncode(&vCtrl,&vInternal);
```

5.2.2. smtSetVideoStatus()/smtGetVideoStatus()

Description

set/get status register of video encoder.

```
void
smtSetVideoStatus(
    VEncStatus *vStatus
);
void smtGetVideoStatus(
    VEncStatus *vStatus
);
```

Parameters

vStatus

video encoder status register 정보를 담고 있는 구조체의 포인터.

Return Value

none

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: videoenc_pre_drv.h

Source: videoenc_pre_drv.c

Library: none

Example Code

```
VEncStatus vStatus;

//set Video Encoder Status Register
vStatus.Enable              = 0x1; // enable video encoder
vStatus.fieldCount          = 0x0;
vStatus.Hcount              = 0x0;
vStatus.Vcount              = 0x0;
smtSetVideoStatus(&vStatus);
```

5.2.3. smtSetVideoSubCarrier()

Description

set the subcarrier frequency setup / subcarrier adfust phase register of video encoder.

```
void smtSetVideoSubCarrier(
    smtUint16 subPhase,
    smtUint32 subReq
);
```

Parameters

subPhase

subcarrier phase value.

subReq

subcarrier frequency setting value.

Return Value

none

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: videoenc_pre_drv.h

Source: videoenc_pre_drv.c

Library: none

Example Code

```
None.
```

5.3. VIF

5.3.1. smtVIFSetMode()/smtVIFGetMode()

Description

set/get control register of external video input.

```
void smtVIFSetMode (VIF_CTRL_STRUCT vif);
void smtVIFGetMode (VIF_CTRL_STRUCT vif);
```

Parameters

vif

VIF control control register 정보를 담고 있는 구조체

Return Value

none.

Remarks

N/A

Requirement

Platform: Requires CT500 firmware platform

Header: vif_pre_drv.h

Source: vif_pre_drv.c

Library: none

Example Code

```
vif.hBlank      = 0;
vif.vBlank      = 0;
vif.field       = 0;
vif.ycOrder     = 2;        // y/cb/y/cr order
vif.startIntEn  = 1;        // Frame start interrupt enable
vif.endIntEn    = 0;        // Frame end interrupt enable
vif.dmaEn       = 1;        // DMA enable

smtVIFSetMode(vif);
```

5.3.2. smtVIFGetIntStatus()

Description

set/get status register of video encoder.

```
smtUint8 smtVIFGetIntStatus (void);
```

Parameters

none

Return Value

VIF start/end frame flag

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: vif_pre_drv.h

Source: vif_pre_drv.c

Library: none

Example Code

```
smtUint8 vifStatus;
vifStatus = smtVIFGetIntStatus();
```

5.3.3. smtVIFSetProperty()/smtVIFGetProperty()

Description

set/get control register of external video input.

```
void smtVIFSetProperty (VIF_PROP_STRUCT vifProp);
void smtVIFGetProperty (VIF_PROP_STRUCT vifProp);
```

Parameters

vifProp

vif property(position, size, address) 정보를 담고 있는 구조체.

Return Value

none

Remarks

none

Requirement

Platform: Requires CT500 firmware platform

Header: vif_pre_drv.h

Source: vif_pre_drv.c

Library: none

Example Code

```
vifProp.vifDmaAddr= FRAME_BASEADDR;
vifProp.vifXPos     = 0;
vifProp.vifYPos     = 0;
vifProp.vifXSize    = 720;
vifProp.vifYSize    = 240;

smtVIFSetProperty(vifProp);
```

6. STORAGE DEVICES

6.1. NAND

6.1.1. smtNANDSetOPT0()

Description

NAND 1st Operation's software interface

```
void smtNANDSetOPT0(NAND_OPT0 *popt0);
```

Parameters

popt0

NAND 1st Operation's 1:1 mapping

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

6.1.2. smtNANDSetOPT1()

Description

NAND 2st Operation's software interface

```
void smtNANDSetOPT1(NAND_OPT1 *popt1);
```

Parameters

popt1

NAND 1st Operation's 1:1 mapping

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

6.1.3. smtNANDSetOPT2()

Description

NAND 3st Operation's software interface

```
void smtNANDSetOPT1(NAND_OPT2 *popt2);
```

Parameters

popt1

NAND 2st Operation's 1:1 mapping

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

6.1.4. smtNANDGetData()/smtNANDSetData()

Description

NAND, NFDATA register software interface

```
void smtNANDSetData(smtUint32 *pdata)
smtUint32 smtUint32 smtNANDGetData(void)
```

Parameters

pdata

data for write NAND

Return Value

Data for read from NAND.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

6.1.5. smtNANDSetCfg()/smtNANDGetCfg()

Description

NAND, NFCONF register software interface

```
void smtNANDSetCfg(NAND_CFG *pconfig)
void smtNANDGetCfg(NAND_CFG *pconfig)
```

Parameters

pconfig

NAND, NFCONF register 1:1 mapping structure

Return Value

none

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

6.1.6. smtNANDSetControl()/smtNANDGetControl()

Description

NAND, NFCTRL register software interface

```
void smtNANDSetControl(NAND_CTRL *pcontrol)
void smtNANDGetControl(NAND_CTRL *pcontrol)
```

Parameters

pcontrol

NAND, NFCTRL register 1:1 mapping structure

Return Value

none

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

6.1.7. smtNANDSetStatus()/smtNANDGetStatus()

Description

NAND, NFSTAT register software interface

```
void smtNANDSetStatus(NAND_STAT *pstatus)
void smtNANDGetStatus(NAND_STAT *pstatus)
```

Parameters

pstatus

NAND, NFSTAT register 1:1 mapping structure

Return Value

none

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

6.1.8. smtNANDGetFIFOStatus()

Description

NAND, NFFIFOSTAT register software interface

```
void smtNANDGetFIFOStatus(NAND_FIFOSTAT *pfifoStatus)
```

Parameters

pstatus

NAND, NFFIFOSTAT register 1:1 mapping structure

Return Value

none

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

6.1.9. smtNANDGetECC()

Description

NAND, ECCSECTOR0~ECCSECTOR8, SECCSECTOR0~SECCSECTOR3 register software interface

```
void smtNANDGetECC(smtUint8 eccIdx, smtUint32 *pecc, smtUint32 *psecc)
```

Parameters

eccIdx

wanted Main/Spare ECC number

pecc

smtUint32 pointer for Main ECC, if NULL no operation for this pointer

psecc

smtUint32 pointer for Spare ECC, if NULL no operation for this pointer

Return Value

none

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: nand_pre_drv.h

Source: nand _pre_drv.c

Library: none

Example Code

```
Refer to test code (peri\storage\nand.c)
```

6.2. PSRAM

6.2.1. smtPSRAMInit()

Description

PSRAMCON, PSRAMTCON, PSRAMTOUT register control function.

```
void smtPSRAMInit (smtUint32 ToutValue)
```

Parameters

ToutValue

Timeout Value

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: psram_pre_drv.h

Source: psram_pre_drv.c

Library: none

Example Code

```
none
```

6.3. SRAM

6.3.1. smtSMCGetControl()/smtSMCSetControl()

Description

ESMC_B0_CON, ESMC_B1_CON, ESMC_B2_CON, ESMC_B3_CON register software interface

```
void smtSMCGetControl(smtUint8 bank, SMC_CTRL *pcontrol)
void smtSMCSetControl(smtUint8 bank, SMC_CTRL *pcontrol)
```

Parameters

bank

ESMC_B0_CON, ESMC_B1_CON, ESMC_B2_CON, ESMC_B3_CON register selector, i.e. 0 means ESMC_B0_CON.

pcontrol

ESMC_B[0-3]_CON register's 1:1 mapping layer

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: smc_pre_drv.h

Source: smc_pre_drv.c

Library: none

Example Code

```
none
```

6.3.2. smtSMCGetControl()/smtSMCSetControl()

Description

ESMC_B0_CON, ESMC_B1_CON, ESMC_B2_CON, ESMC_B3_CON register software interface

```
void smtSMCGetControl(smtUint8 bank, SMC_CTRL *pcontrol)
void smtSMCSetControl(smtUint8 bank, SMC_CTRL *pcontrol)
```

Parameters

bank

ESMC_B0_CON, ESMC_B1_CON, ESMC_B2_CON, ESMC_B3_CON register selector, i.e. 0 means ESMC_B0_CON.

pcontrol

ESMC_B[0-3]_CON register's 1:1 mapping layer

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: smc_pre_drv.h

Source: smc_pre_drv.c

Library: none

Example Code

```
none
```

6.4. DDR-SDRAM

6.4.1. smtDMCSetControl()/smtDMCGetControl()

Description

DDRCON register software interface

```
void smtDMCSetControl(DMC_CTRL *pcontrol)
void smtDMCGetControl(DMC_CTRL *pcontrol)
```

Parameters

pcontrol

DDRCON register's 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: dmc_pre_drv.h

Source: dmc_pre_drv.c

Library: none

Example Code

```
none
```

6.4.2. smtDMCGetTimingControl()/smtDMCSetTimingControl()

Description

DDRTCON register software interface

```
void smtDMCSetTimingControl(DMC_TIMING *ptiming)
void smtDMCGetTimingControl(DMC_TIMING *ptiming)
```

Parameters

ptiming

DDRTCON register's 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: dmc_pre_drv.h

Source: dmc_pre_drv.c

Library: none

Example Code

```
none
```

6.4.3. smtDMCGetPowerControl()/smtDMCSetPowerControl()

Description

DDRPCON register software interface

```
void smtDMCSetPowerControl(DMC_POWER *ppower)
void smtDMCGetPowerControl(DMC_POWER *ppower)
```

Parameters

ppower

DDRPCON register's 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: dmc_pre_drv.h

Source: dmc_pre_drv.c

Library: none

Example Code

```
none
```

6.4.4. smtDMCGetRefreshControl()/smtDMCSetRefreshControl()

Description

DDRREF register software interface

```
void smtDMCSetRefreshControl(DMC_REFRESH *prefresh)
void smtDMCGetRefreshControl(DMC_REFRESH *prefresh)
```

Parameters

prefresh

DDRREF register's 1:1 mapping structure

Return Value

none.

Remarks

none.

Requirement

Platform: Requires CT500 firmware platform

Header: dmc_pre_drv.h

Source: dmc_pre_drv.c

Library: none

Example Code

```
none
```